Abu Dhabi City Campus Daily/Weekly Planner

Week 11: 9th – 12th November 2026

Subject: Mathematics Pre-AP Geometry with Statistics **Unit 2: Tools and Techniques of Geometric Measurement** **Grade: 9**

Lessons	Learning targets	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1: **Unit 1 Performance Task – Designing a Study**	I can explain the purpose, stages and success criteria of the Designing a Study performance task and identify what makes a research question suitable for statistical investigation.	**2.1** question-analysis template, vocabulary bank and two guided question choices; Core—compare and justify a suitable survey or simulation question; Extension—propose an original question and defend its complexity, feasibility and data requirements.	Unit 2 Key Concept 2.1 Student Handouts 2.1.B and 2.1.C Pre-AP Geometry, Lesson 2.1.1 Describe and correctly label a line, ray, or line segment. Pre-AP Geometry, Lesson 2.1.2 Describe and correctly label an angle. Pre-AP Geometry, Lesson 2.1.4 Measure an angle. Pre-AP Geometry, Lesson 2.1.5 Prove whether two or more line segments are congruent. Pre-AP Geometry, Lesson 2.1.6 Prove whether two or more angles are congruent.
Lesson 2: **Performance Task** **Part A: Research Question** **Part B: Design**	I can select or refine a research question and design a survey or simulation that can generate appropriate data while identifying and addressing potential sources of bias.	**2.3** Teacher conference and checklist aligned to rubric: research question explained; instrument/procedure can answer it; potential bias identified and addressed. Students submit a draft plan and receive one strength and one actionable next step.	
Lesson 3: **Measuring Line Segments**	I can measure line segments using standard and informal units and determine whether two or more segments are congruent by comparing their lengths or matching them through a rigid motion.	**2.1** Individual check: identify a congruent segment pair, state the evidence and distinguish the geometric object (segment) from its numerical measure (length). Check measurement accuracy, notation and justification.	
Lesson 4: **Measuring Congruent Angles**	I can measure angles and determine whether two or more angles are congruent by comparing their measures or using a rigid motion to make them coincide.	**2.3** Support—translucent tracing paper, labelled protractor guide, colour-coded vertices and sentence frames; Core—match, measure and justify congruent angles; Extension—explain why angles with different side lengths may still be congruent and connect the measurement and rigid-motion definitions.	
Homework	Students need to complete Part C (Data collection and representation) of the performance task assigned on Teams and College Board. Student need to complete handout 2.1 B and C assigned on College Board.		
Teacher's notes / Reminders	Students should bring a device, scientific calculator and notebook and have access to the PPTs.		

Technology Integration	Microsoft Teams and OneNote; projector; graphing utility/calculators; performance-task planning templates; College Board handouts and rubric; rulers, unmarked straightedges, index cards, string, compasses, patty paper/translucent paper and protractors.
Cross-curricular links	Statistics/Social Sciences: designing surveys and interpreting sample responses. Design & Technology: precision in measurement and comparing geometric objects.
External Assessment Integration	College Board Pre-AP Geometry with Statistics
Wellbeing/ ATL Skills	**Thinking:** formulate research questions, select methods, identify bias, compare measurements and justify congruence. **Communication:** explain design decisions, present evidence, use correct segment/angle notation and justify conclusions. **Self-management:** plan project steps, organise notes, use tools carefully and meet checkpoints. **Social:** collaborate in pairs, discuss methods and give constructive feedback.
Standards	Unit 1 Performance Task: Designing a Study – Part A Research Question; Part B Design; consider sample size, appropriate data collection and potential sampling/response bias. Unit 2: Tools and Techniques of Geometric Measurement – Key Concept 2.1. 2.1.3 Measure a line segment (2.1.3a–b). 2.1.4 Measure an angle (2.1.4a). 2.1.5 Prove whether line segments are congruent (2.1.5a–b). 2.1.6 Prove whether angles are congruent (2.1.6a–b).
Inquiry/ TOK Question	How can the way we design a study affect the evidence and conclusions we obtain, and how can measurement and movement help us decide whether two geometric objects are congruent?
Success Criteria (Measurable)	**Lesson 1: Introducing the Unit 1 Performance Task – Designing a Study** • **All students must** identify the task stages and explain the purpose of a suitable research question. • **Most students should** compare survey and simulation options and justify which type is appropriate for a given question. • **Some students could** propose an original research question and defend its suitability, feasibility and data requirements. **Lesson 2: Performance Task Workshop – Parts A and B** • **All students must** select a research question and draft a survey or simulation design. • **Most students should** explain how the instrument/procedure will generate data to answer the question and identify potential bias. • **Some students could** evaluate alternative designs and justify improvements that reduce bias while preserving the research purpose.

	Lesson 3: Measuring Line Segments • **All students must** measure or compare line segments using an appropriate tool and identify equal lengths. • **Most students should** justify whether segments are congruent using equal measures and correct notation. • **Some students could** compare segments using informal or indirect measurement and explain why orientation and position do not affect congruence. **Lesson 4: Measuring Congruent Angles** • **All students must** identify and measure angles using tracing/matching or a protractor. • **Most students should** justify whether angles are congruent using equal measures or a rigid-motion match. • **Some students could** explain why angles with different side lengths can be congruent and connect the measure-based and rigid-motion definitions.
National Identity Domain	Area: Citizenship
	Implementation: students design a study that considers the population being represented and uses neutral, respectful questions.
	Area: Responsible decision-making. Implementation: students identify limitations and bias in their study design and use precise measurement and evidence when making geometric claims.